

Chapter 5

Analysis of Variance

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Learning Objectives

When we have completed this chapter, we will be able to:

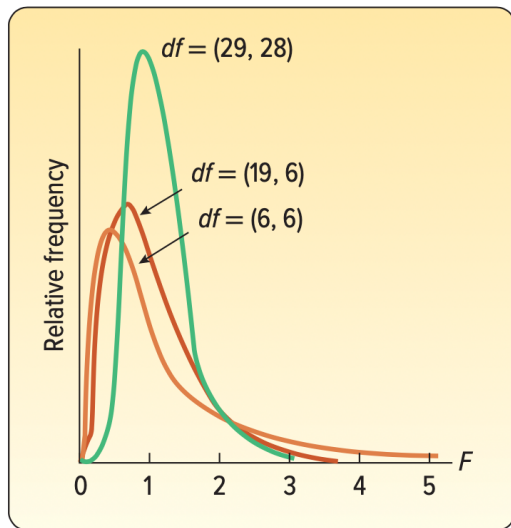


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- Apply the F distribution to test a hypothesis that two population variances are equal.
- Use ANOVA to test a hypothesis that three or more population means are equal.

About F distribution

It was named to honor Sir Ronald Fisher, one of the founders of modern-day statistics.



- F distribution is continuous
- F-statistic cannot be negative.
- F distribution is positively skewed.
- F distribution is asymptotic.

Example 1.0

Lammers Limos offers limousine service from Government Center in downtown Toledo, Ohio, to Metro Airport in Detroit. Sean Lammers, president of the company, is considering two routes. One route is via U.S. Route 25 and the other via Interstate 75.

He wants to study the time it takes to drive to the airport using each route and then compare the results. He collected the following sample data, reported in minutes.

Using the 0.10 significance level, is there a difference in the variation in the driving times for the two routes?

U.S. Route 25	Interstate 75
52	59
67	60
56	61
45	51
70	56
54	63
64	57
	65

Solution 1.0

U.S. Route 25

$$\bar{x}_1 = \frac{408}{7} = 58.29$$

$$s_1 = \sqrt{\frac{485.43}{6}} = 8.9947$$

Interstate 75

$$\bar{x}_2 = \frac{472}{8} = 59.00$$

$$s_2 = \sqrt{\frac{134}{7}} = 4.3753$$

F-Test for Equality of Variances (Steps 1–4)

Step 1: State the hypotheses. The test is **two-tailed** because we are testing for a difference in variation, not whether one route has more variation.

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

Step 2: Select the significance level.

$$\alpha = 0.10$$

$$\alpha/2 = 0.05$$

Step 3: Select the test statistic. The test statistic follows an **F distribution**.

$$df_1 = n_1 - 1 = 6, \quad df_2 = n_2 - 1 = 7$$

Step 4: Determine the critical value using the F table.

$$F_{\text{critical}} = 3.87$$

Decision rule: Reject H_0 if $F > 3.87$.

Test Statistics

Step 5: Compute the ratio of the two sample variances, determine the value of the test statistic, and make a decision regarding the null hypothesis.

$$F = \frac{s_1^2}{s_2^2} = \frac{(8.9947)^2}{(4.3753)^2} = 4.23$$

Decision: Reject H_0 . We conclude that there is a **difference in the variation** of the travel times between the two routes.

As a result, Mr. Lammers should take this difference in variability into account when making scheduling decisions.

TABLE 12-1 Critical Values of the F Distribution, $\alpha = .05$

Degrees of Freedom for Denominator	Degrees of Freedom for Numerator			
	5	6	7	8
1	230	234	237	239
2	19.3	19.3	19.4	19.4
3	9.01	8.94	8.89	8.85
4	6.26	6.16	6.09	6.04
5	5.05	4.95	4.88	4.82
6	4.39	4.28	4.21	4.15
7	3.97	3.87	3.79	3.73
8	3.69	3.58	3.50	3.44
9	3.48	3.37	3.29	3.23
10	3.33	3.22	3.14	3.07

Excel software

The Excel software has a procedure to perform a test of variances.

J6							
	A	B	C	D	E	F	G
1	A (U.S. 25)	B (Interstate 75)					
2	52	59		F-Test Two-Sample for Variances			
3	67	60					
4	56	61			Variable 1	Variable 2	
5	45	51		Mean	58.2857143	59	
6	70	56		Variance	80.9047619	19.1428571	
7	54	63		Observations	7	8	
8	64	57		df	6	7	
9		65		F	4.22636816		
10				P(F<=f) one-t	0.04039661		
11				F Critical one	2.82739227		
12							
13							

The F of 4.23 is greater than the critical value of 3.87. Also, the p-value is less than 0.05. We conclude the variance of travel times on U.S. 25 is greater than the variance of travel times on I-75.

F Distribution: Key Concepts

Analysis of Variance (ANOVA)

A technique used to test simultaneously whether the means of several populations are equal. The test statistic follows an **F distribution**.

Test Statistic for Comparing Two Variances

$$F = \frac{s_1^2}{s_2^2}$$

ANOVA: Analysis of Variance

The **F distribution** is used to perform a wide variety of hypothesis tests. When testing the equality of three or more population means, the **analysis of variance (ANOVA)** technique is used, and the **F-statistic** serves as the test statistic.

ANOVA Assumptions

Using ANOVA to test the equality of three or more population means requires that the following assumptions are satisfied:

- 1 The populations follow a normal distribution.
- 2 The populations have equal standard deviations (σ).
- 3 The populations are independent.

Example 1.1

Use ANOVA to test a hypothesis that three or more population means are equal.

Dara manages a regional financial

center. She wishes to compare the productivity, as measured by the number of customers served, among three employees. Four days are randomly selected, and the number of customers served by each employee is recorded. Is there a difference in the mean number of customers served?

Wolfe	White	Korosa
55	66	47
54	76	51
59	67	46
56	71	48

Comparison of Figures

Chart 12–1 illustrates how the populations would appear if there were a difference in the treatment means. Note that the populations follow the normal distribution and the variation in each population is the same. However, the means are not the same.

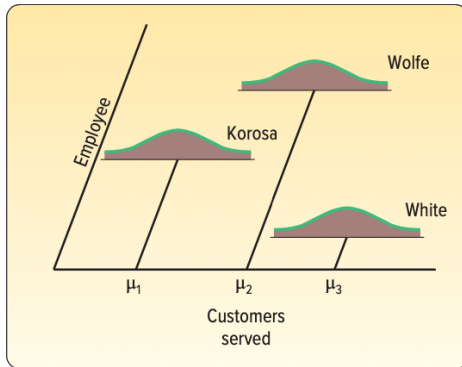


CHART 12–1 Case Where Treatment Means Are Different

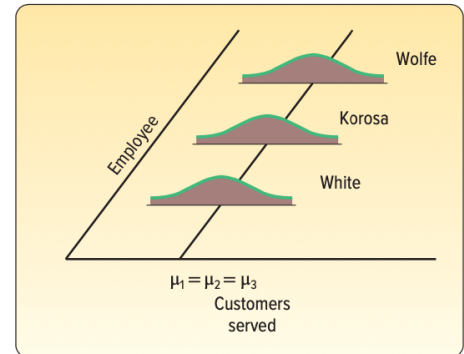


CHART 12–2 Case Where Treatment Means Are the Same

Airline Customer Satisfaction (ANOVA Example)

The airline industry monitors and tracks customer satisfaction. Information is collected on issues such as overbooking and baggage handling. Brunner Marketing Research Inc. surveyed passengers from four airlines regarding their satisfaction with a recent flight. Responses were scored as follows: Excellent = 4, Good = 3, etc.

The total score represents the overall satisfaction level. A higher score indicates greater satisfaction, with a maximum possible score of 100.

Northern WTA Pocono Branson			
94	75	70	68
90	68	73	70
85	77	76	72
80	83	78	65
	88	80	74
		68	65
		65	

Question: Is there a difference in the mean satisfaction level among the four airlines? Use the $\alpha = 0.01$ significance level.

One-Way ANOVA: Hypothesis Testing Procedure

We use the **six-step hypothesis-testing procedure**.

Step 1: Hypotheses

The null hypothesis states that the mean satisfaction scores are the same for the four airlines. The alternative hypothesis states that **not all mean scores are equal** (i.e., at least two means differ).

Step 2: Select the significance level

$$\alpha = 0.01$$

Step 3: Determine the test statistic The test statistic follows an **F distribution**.

$$H_0 : \mu_N = \mu_W = \mu_P = \mu_B$$

vs

$$H_1 : \text{Not all means are equal}$$

Step 4: Decision Rule

The critical value of the F-statistic is obtained from the F table using:

$$df_{\text{numerator}} = k - 1$$

$$df_{\text{denominator}} = N - k$$

Reject H_0 if the computed F-statistic exceeds the critical value.

ANOVA: Degrees of Freedom and Decision Rule

The degrees of freedom in the denominator equal the total number of observations minus the number of treatments.

$$df_{\text{numerator}} = k - 1 = 4 - 1 = 3$$

For this problem:

- Number of treatments: $k = 4$
- Total observations: $n = 22$

$$df_{\text{denominator}} = n - k = 22 - 4 = 18$$

Using the F table for $\alpha = 0.01$, we locate the critical value using the numerator and denominator degrees of freedom.

$$F_{\text{critical}} = 5.09$$

Decision rule: Reject H_0 if the computed F-statistic exceeds the critical value.

Reject H_0 if $F > 5.09$

ANOVA Table and Sum of Squares

ANOVA Table Structure

Source	SS	df	MS	F
Treatments	SST	$k - 1$	MST	MST/MSE
Error	SSE	$n - k$	MSE	
Total	SS total	$n - 1$		

SST measures variation due to treatments, SSE measures variation within treatments.

We begin by computing the total sum of squares:

$$SS_{\text{total}} = \sum (x - \bar{x}_G)^2$$

where $\bar{x}_G$ is the grand mean.

Next, compute the sum of squares error:

$$SSE = \sum (x - \bar{x}_c)^2$$

ANOVA: Grand Mean and SS Total

Sample Data Summary

	Northern	WTA	Pocono	Branson	Total
Column total	349	391	510	414	1,664
n	4	5	7	6	22
Mean	87.25	78.20	72.86	69.00	75.64

The **grand mean** is computed using all observations:

$$\bar{x}_G = \frac{1664}{22} = 75.64$$

Total Sum of Squares

We compute the deviation of each observation from the grand mean, square each deviation, and sum over all observations.

ANOVA: Deviations and Error Sum of Squares

Deviations from the Grand Mean

Northern	WTA	Pocono	Branson
18.36	−0.64	−5.64	−7.64
14.36	−7.64	−2.64	−5.64
9.36	1.36	0.36	−3.64
4.36	7.36	2.36	−10.64
	12.36	4.36	−1.64
		−7.64	−10.64
		−10.64	

Squared Deviations

Each deviation is squared and summed to obtain SS_{total} .

$$\begin{aligned}SS_{\text{total}} &= \sum (x - \bar{x}_G)^2 \\&= (94 - 75.64)^2 + \dots \\&= (18.36)^2 + \dots \\&= 337.09 + \dots = 1485.10\end{aligned}$$

ANOVA: Deviations from Treatment Means and SSE

Deviations from Treatment Means

Northern	WTA	Pocono	Branson
6.75	-3.20	-2.86	-1
2.75	-10.20	0.14	1
-2.25	-1.20	3.14	3
-7.25	4.80	5.14	-4
	9.80	7.14	5
		-4.86	-4
		-7.86	

Next Step: Error Sum of Squares

To compute SSE , we measure deviations from each **treatment mean**:

$$SSE = \sum (x - \bar{x}_c)^2$$

where $\bar{x}_c$ is the mean of treatment c (e.g., $\bar{x}_N = 349/4 = 87.25$).

ANOVA: SSE, SST, and Test Statistic

Error Sum of Squares

Squaring each deviation and summing across all observations gives:

$$SSE = \sum (x - \bar{x}_c)^2 = 594.41$$

Sum of Squares for Treatments

$$SST = SS_{\text{total}} - SSE$$

$$SST = 1485.10 - 594.41 = 890.69$$

The mean square values are:

$$MST = \frac{SST}{k - 1}, \quad MSE = \frac{SSE}{n - k}$$

ANOVA Table

Source	SS	df	MS	<i>F</i>
Treatments	890.69	3	296.90	8.99
Error	594.41	18	33.02	
Total	1485.10	21		

$$F = \frac{MST}{MSE} = 8.99$$

Step 6: Decision and Interpretation

Decision

The computed F-statistic is:

$$F = 8.99$$

The critical value is:

$$F_{\text{critical}} = 5.09$$

Since

$$8.99 > 5.09,$$

we **reject the null hypothesis**.

Interpretation

We conclude that the population means are **not all equal**. The ANOVA results indicate that at least one pair of mean satisfaction scores differs among the four airlines.

However, ANOVA does not identify *which* airlines differ or which airline has the highest or lowest satisfaction. Additional post-hoc comparison methods are required to determine where the differences occur.

Excel software

The Excel software has a procedure to perform ANOVA:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Northern	WTA	Pocono	Branson								
2	94	75	70	68		Anova: Single Factor						
3	90	68	73	70								
4	85	77	76	72		SUMMARY						
5	80	83	78	65		<i>Groups</i>	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>		
6		88	80	74		Northern	4	349	87.25	36.9166667		
7			68	65		WTA	5	391	78.2	58.7		
8			65			Pocono	7	510	72.8571429	30.1428571		
9						Branson	6	414	69	13.6		
10												
11												
12						ANOVA						
13						<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
14						Between Groups	890.6837662	3	296.894589	8.9906433	0.00074277	5.09188952
15						Within Groups	594.4071429	18	33.022619			
16												
17						Total	1485.090909	21				
18												

The F of 8.99 is greater than the critical value of 5.09. Also, the p-value is less than 0.01, so the null hypothesis is rejected.

One-Way ANOVA Results

ANOVA

Source	SS	df	MS	F
Between Groups	6.90	2	3.45	5.15
Within Groups	12.04	18	0.67	
Total	18.94	20		

Questions

- a. How many treatments are in the study?
- b. What is the total sample size?
- c. What is the critical value of F ?
- d. Write the null and alternative hypotheses.
- e. What is your decision regarding H_0 ?
- f. Can we conclude any treatment means differ?